
Researching Historical-Biographical Information in Digital Scholarly Editions through Event-Oriented Affiliation Networks

Timo Frühwirth^{*1}

¹Austrian Centre for Digital Humanities, Austrian Academy of Sciences (ACDH) – Austria

Abstract

Auden in Austria (Digital)

British-American poet W. H. Auden (1907-1973) was one of the most acclaimed writers of the twentieth century. From 1958 to 1973, he lived and worked in Austria, where he wrote most of his late poetry. However, whereas the 'British' and 'American Auden' have been thoroughly investigated, Auden's Austrian period has long remained understudied. In the context of an emerging field of Austrian Auden studies, Auden in Austria Digital (AAD, Austrian Academy of Sciences) provides open access to all Auden documents in Austrian archives – literary papers, but also correspondence and legal records – through a digital scholarly edition (Mayer et al. 2025).

Formalizing Historical-Biographical Information as TEI 'Event'

The two core types of research objectives in the AAD project, (1) philological and (2) historical-biographical, correspond to two key data requirements: (1) machine-readable full texts, and (2) machine-readable data created from what Vogeler (2019) terms "facts asserted by the transcription". Beyond TEI/XML textual markup, AAD formally models relevant historical-biographical information through the TEI 'event' element, which interconnects events with person and organization entities. In the AAD edition, such event modeling includes not only well-documented historical occurrences, but also small-scale interactions in which Auden participated – meetings and visits, telephone calls, previous correspondence referred to in Auden's letters – of which the edition documents are typically the sole source.

From TEI 'listEvent' to Event-Oriented Affiliation Network

By formalizing event co-affiliation, the full dataset (available through the TEI 'listEvent') makes these interpersonal relations accessible for computational analysis of the network that extends around the 'Austrian Auden'. In this affiliation network, participation creates edges between person/organization nodes and event nodes in a bipartite graph; through unipartite projection, actors can be directly connected to the central actor in a derived ego network. Such an approach takes advantage of TEI event data and avoids the need for standoff relationship modeling that is outside standard editorial practice; data natively created in digital scholarly editing thus become the basis for network analysis of archival source materials. We propose such an event-oriented affiliation-network approach as a practicable method for interconnecting digital editing with network analysis: while recent research demonstrates the

^{*}Speaker

potential of event-based affiliation networks (Levchenko 2025), this paper centers on operationalizing such network research within a TEI/XML framework in a reusable way.

Workflow, Challenges, and Scope

While AAD data creation is work in progress, the key components of an XSLT-and-Python post-processing workflow have been assembled and are openly available (Frühwirth 2025a and 2025b). With regard to the data pipeline, this paper addresses methodological challenges both upstream (during TEI data creation) and downstream (during the further processing of TEI data and network building). The paper concludes by considering both limitations and potentials of an affiliation-network approach in the context of literary-history and biographical research: while such networks cannot capture the quality of individual social contacts, they can map the interpersonal configurations in which an actor moves – such as the social and artistic milieus of the ‘Austrian Auden’.

References

Frühwirth, Timo. 2025a. "affiliation-builder." Python Package. Version 0.2.0. Python Package Index. <https://pypi.org/project/affiliation-builder>

Frühwirth, Timo. 2025b. listEvent-to-json.xsl. GitHub. <https://github.com/timofruehwirth/event-network/tree/main/xslt>

Levchenko, Maria. 2025. "Computational Analysis of Literary Communities: Event-Based Social Network Study of St. Petersburg 1999-2019." *Journal of Computational Literary Studies* 4 (1). <https://doi.org/10.48694/jcls.4217>

Mayer, Sandra, Timo Frühwirth, Edward Mendelson, Peter Andorfer, Daniel Elsner, and Angelika Hechtel, eds. 2025. Auden in Austria Digital. Version 0.1.0. <https://auden.acdh.oeaw.ac.at>
Vogeler, Georg. 2019. "The 'Assertive Edition': On the Consequences of Digital Methods in Scholarly Editing for Historians." *International Journal of Digital Humanities* 1: 309–322. <https://doi.org/10.1007/s42803-019-00025-5>

Keywords: DSE, edition, TEI, XML, affiliation network, Python, NetworkX, XSLT, bipartite, event, ego network, 20th century, Austria